Focus Session Outline

Guided Discussion

DOMMx – Data Operating Model & Maturity

An Integrated Operating Model for Assessment, Governance Execution, and Continuous Capacity Improvement

LEONARDO GUERREIRO, JOSÉ MARTINS, MARIA DO ROSÁRIO BERNARDO, HENRIQUE MAMEDE, AND FREDERICO BRANCO

Executive Summary

This paper presents the roadmap of the Focus Session designed for the academic and operational validation of DOMMx – Data Operating Model & Maturity, in the context of PhD research. The Focus Session is a structured qualitative instrument designed to collect empirical evidence to assess the conceptual coherence, methodological consistency and practical applicability of the proposed model.

DOMMx emerges as a response to the gaps identified in the Systematic Review of the Literature (SLR), particularly with regard to the absence of end-to-end operating models capable of linking, in an explicit and traceable way, maturity assessment, decision-making and practical execution of data governance. The Focus Session does not aim to confirm or reproduce existing literature, but rather to empirically assess whether the proposed operating model adequately responds to these gaps.

The script structures the session as a guided debate, supported by a prior maturity questionnaire, allowing participants to understand in a concrete way how DOMMx works before the discussion. The debate is organised into thematic blocks covering the conceptual clarity of the model, its internal coherence, operational feasibility in real-world contexts and the main risks and assumptions associated with its adoption.

From a methodological point of view, the Focus Session is aligned with consolidated qualitative research practices in Information Systems and Software Engineering, according to the guidelines proposed by Kitchenham for complementary studies to SLRs. The protocol for recording, analyzing and using the results ensures traceability between the participants' contributions, the SLR research questions and the scientific validation of the model.

This document is intended to guide the participants of the Focus Session, clarifying objectives, structure, roles, reference materials and evaluation criteria, as well as to serve as a formal methodological artifact to support the scientific validity of the research.

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1.0 GENERAL FRAMEWORK OF THE FOCUS SESSION

1.1 Purpose of the Focus Session

The Focus Session aims to carry out a critical, informed and structured evaluation of DOMMx, bringing together participants with practical and academic experience, in order to validate the conceptual coherence, methodological consistency and practical applicability of the model, in the context of a doctoral thesis.
The session aims to generate qualitative empirical evidence to assess whether DOMMx responds to the gaps identified in the Systematic Literature Review (SLR), namely with regard to the operationalization of data governance, the evolution of organizational maturity and the link between diagnosis, decision and execution.

1.2 Role of the Focus Session in the Methodology of the Thesis

The Focus Session works as a qualitative instrument complementary to the SLR, not with the purpose of confirming or reproducing the existing literature, but rather to empirically evaluate the proposed solution in the light of this literature.
Its use is aligned with formal qualitative research methodologies recommended for the validation of models, artifacts and theoretical proposals in Information Systems and Software Engineering.

2.0 PARTICIPANT PROFILE AND SESSION ORGANIZATION

2.1 Profile of two Participants

The session has approximately 10 participants, distributed as follows:

- 5 participants with active involvement in the debate
- 5 participants with observation paper and scientific record

Participants have professional and/or academic experience in areas related to data, data governance, data architecture, organizational management or scientific research.

2.2 Roles during the session

2.2.1 Moderator

The moderator is responsible for leading the session, presenting the mediated questions, ensuring a balance of participation, managing time and ensuring that the debate remains aligned with the scientific objectives of the research.

2.2.2 Active Participants

Active participants contribute with critical analysis, conceptual insights and practical experience, evaluating DOMMx from both theoretical and operational perspectives.

2.2.3 Observers and Registrars

Observers follow the session with a focus on identifying consensus patterns, divergences, implicit assumptions, and critical points relevant for further analysis.

3.0 FOCUS SESSION REFERENCE MATERIALS

3.1 Materials Previously Submitted

Participants receive the following materials in advance:

- DOMMx Executive Summary
- Prior Maturity Questionnaire (Typeform)
- Overview of the DOMMx operating model framework
- Systematic Literature Review (SLR), limited to an overview of its structure, including:
- Objectives of the review
- Research Questions
- Top Analytical Categories

Participants are not expected to read the SLR in its entirety or the full model, but to understand the theoretical framework, the logic of the model, and the relationship between maturity, decision, and execution.

3.2 Role of the Prior Questionnaire

The preliminary questionnaire aims to align the understanding of the participants, demonstrate in a reduced way the functioning of the DOMMx and serve as a concrete basis for the debate during the Focus Session.

4.0 FORMAT AND CONDITIONS OF THE SESSION

4.1 Modality

The Focus Session will be held online.

4.2 Total Duration

The total duration of the session is 90 minutes.

4.3 Session Registration

The session will be recorded in full audio and video. Subsequently, a partial transcription will be carried out guided by the thematic blocks of the script.

5.0 STRUCTURE OF THE DEBATE AND MEDIATED ISSUES

5.1 Opening of the Session *(5 minutes)*

The moderator presents the objective of the session, the framing of DOMMx in the context of the thesis and explains the dynamics of the debate.

5.2 Cognitive Warm-Up *(10 minutes)*

Was the relationship between the answers to the previous questionnaire, the maturity levels identified, the expected deliverables, and the actions triggered by DOMMx clear and understandable?

5.3 Block 1 – Conceptual Clarity and Positioning *(15 minutes)*

Was DOMMx clearly perceived as an end-to-end operating model, and not just as a generic framework or an isolated maturity assessment?

5.4 Block 2 – Internal Coherence and Operational Logic *(15 minutes)*

Did the DOMMx structure, by linking assessment, maturity levels, actions, procedures, and evidence, present logical coherence and internal consistency?

5.5 Block 3 – Practical Applicability and Operational Feasibility *(20 minutes)*

Does DOMMx support in a practical way the prioritization and structuring of governance initiatives, whether by domains, organizational areas or incremental initiatives?

5.6 Block 4 – Limitations, Risks and Assumptions *(15 minutes)*

What are the main limitations, organizational assumptions, and risks associated with adopting DOMMx?

5.7 Closing and Consolidation *(10 minutes)*

To what extent does DOMMx contribute to filling the gaps identified in the literature on data governance?

6.0 PROTOCOL FOR REGISTRATION AND SCIENTIFIC ANALYSIS

6.1 Type of Evidence Collected

The empirical evidence collected includes:

- Audiovisual recording of the session
- Verbal contributions from participants
- Observers' structured notes

6.2 After-Analysis Strategy

The analysis will be conducted through thematic qualitative analysis, with coding guided by the blocks of the script and explicit mapping of evidence to SLR issues.

7.0 USE OF FOCUS SESSION RESULTS

The results of the Focus Session will be used to empirically validate DOMMx, support model refinement when necessary, and sustain discussion of thesis results.

8.0 FOCUS SESSION EVALUATION PROTOCOL

The Focus Session is considered valid as a scientific tool if it allows the collection of qualified empirical evidence, the triangulation between the SLR, the prior questionnaire and the structured debate, and the identification of patterns, convergences and divergences in the perceptions of the participants.
The data collected will be analyzed in a systematic way, ensuring traceability between empirical evidence, research questions and study conclusions.

9.0 ACADEMIC AND METHODOLOGICAL FRAMEWORK:

"*ALIGNMENT BETWEEN THE FOCUS SESSION AND THE SLR*".

This chapter presents the academic and methodological framework that ensures the alignment between the SLR and the Focus Session conducted for validation of the DOMMx operating model.
In accordance with the methodological guidelines proposed by Kitchenham, the Focus Session does not aim to reproduce or confirm results from the literature, but rather to empirically assess whether the proposed model responds to the gaps identified in the SLR, particularly with regard to the operationalization of data governance and the evolution of organizational maturity.

9.1 Methodological Framework of the Focus Session

The Focus Session is designed as a guided, structured and recorded debate, involving participants with practical and academic experience in data, governance and organizational management.

9.2 Mapping Between SLR Issues and the Focus Session Roadmap

The structure of the script was designed to explicitly answer the research questions defined in the SLR, ensuring direct alignment between theory and empirical validation, as summarized in Table 1.

SLR Issue	**Roadmap Blocks**	**Evidence Assessed**
RQ0 – How can a structured master data management framework improve data governance maturity?	Block 2, Block 3, Closure	Link between maturity, decision and execution
RQ1 – What are the most relevant concepts, standards, and models used as references in data management and governance?	Block 1, Block 2	Conceptual and theoretical coherence of the model
RQ2 – Are there standardized organizational data governance frameworks?	Block 1, Block 4	Positioning against existing frameworks
RQ3 – What are the most widely used maturity models in data governance?	Heating, Block 2, Block 4	Clarity and proportionality of maturity
RQ4 – What are the key antecedents and impacts on governance maturity?	Block 3, Block 4, Closure	Organizational background and impacts

Table 1

9.3 Theoretical Basis by Script Block

Each block of the script is based on references identified in the SLR, ensuring scientific coherence between the literature, the proposed model, and the empirical validation instrument.

9.3.1 Block 1 – Conceptual Clarity and Positioning of DOMMx

This block is based on the literature that analyzes the conceptual nature of data governance frameworks and the absence of end-to-end operating models capable of linking theory and practice.

References:
Abraham, R., Schneider, J., & vom Brocke, J. (2019). Data governance: A conceptual framework, structured review, and research agenda. International Journal of Information Management, 49, 424–438.
Otto, B. (2011). Organizing data governance: Findings from the telecommunications industry. Journal of Enterprise Information Management, 24(6), 542–561.
Legner, C., Pentek, T., & Otto, B. (2020). Accumulative development of data governance capabilities. Information Systems Management, 37(3), 229–245.

9.3.2 Block 2 – Internal Coherence and Operational Logic of the Model

This block is based on studies that address governance mechanisms, integration between domains, organizational scope, and the articulation between diagnosis, decision, and execution.

References:
Abraham, R., Schneider, J., & vom Brocke, J. (2019).
Janssen, M., Weerakkody, V., Ismagilova, E., Sivarajah, U., & Irani, Z. (2020). A framework for analysing big data governance. Government Information Quarterly, 37(3), 101493.
Nielsen, J. A. (2017). Practical issues in data governance implementation. Information Systems Management, 34(1), 24–38.

9.3.3 Block 3 – Practical Applicability and Operational Feasibility

This block is based on literature that emphasizes the need for governance models that are actionable, feasible, and adaptable to different organizational contexts.

References:
Bento, F., Costa, C., & Aparicio, M. (2022). Data governance maturity models: A systematic literature review. Journal of Business Research, 146, 330–345.
Lis, D., Otto, B., & Österle, H. (2023). Antecedents of data governance success. Information & Management, 60(1), 103660.
Ismail, N., et al. (2024). Implementing DAMA-DMBOK in practice. Journal of Data and Information Quality.

9.3.4 Block 4 – Limitations, Risks and Assumptions of the Model

This block is aligned with studies that discuss organizational maturity, critical success factors, culture, executive sponsorship, and risks associated with the adoption of governance models.

References:
Mahanti, R. (2021). Data governance maturity models: A comparative study. International Journal of Information Management, 56, 102226.
Kurniawan, N. B., Suhardi, & Sunindyo, W. D. (2019). Data governance maturity model development. Proceedings of Computer Science, 161, 1140–1147.
Lis, D., Otto, B., & Österle, H. (2023).

10.0 FINAL CONSIDERATIONS

This chapter summarizes the role of the Focus Session in the global context of research, highlighting its contribution to the scientific validation of DOMMx, the methodological robustness of the study and the strengthening of the link between systematic review of the literature, proposed operational model and empirical evidence.